
Analysis and Performance Evaluation of Li-Ion Batteries Using NASA Battery Data Set

TANISH PATEL *, STUDENT, FAU ERLANGEN

¹Friedrich-Alexander-Universität, 91054 Erlangen, Germany

CORRESPONDING AUTHOR: TANISH PATEL (e-mail: tanish.patel@fau.de).

ABSTRACT The report, "Analysis and Performance Evaluation of Li-Ion Batteries Using NASA Battery Data Set," presents a detailed analysis of the performance and characteristics of Li-ion batteries under various operational conditions. Utilizing data from the NASA Prognostics Center of Excellence (PCoE), which includes charge, discharge, and impedance measurements across different temperatures and operational profiles, the study aims to understand battery behavior and identify key performance metrics. The NASA Battery Data Set comprises multiple files representing specific batteries and experimental conditions. Each battery's data includes charge (voltage, current, temperature, time), discharge (voltage, current, temperature, time, capacity), and impedance (current, voltage, impedance values) measurements. Batteries were tested under varying temperatures (ambient, elevated, room, and low temperatures) and different discharge stopping voltages (2.2V, 2.5V, and 2.7V). The charging method used was Constant Current (CC) at 1.5A until 4.2V, followed by Constant Voltage (CV) until the current dropped to 20mA. Discharge was conducted at 4A until the specified stopping voltage.

The experimental procedures aimed to simulate real-world conditions, and Electrochemical Impedance Spectroscopy (EIS) was used for impedance measurement. Results indicate a sharp initial rise in voltage during charging, followed by stabilization in the CV phase, and a gradual decrease during discharge. Current remained constant during the CC phase and decreased in the CV phase. Temperature increased during charging and decreased slightly during discharge, staying within safe operational limits. These findings provide insights into battery performance degradation and overall health, which are crucial for future battery design and application.

INDEX TERMS Battery Behavior, Battery Capacity, Battery Data Analysis, Battery Degradation, Battery Health, Battery Testing Conditions, Charge-Discharge Cycles, Constant Current Charging, Constant Voltage Charging, Electrochemical Impedance Spectroscopy (EIS), Experimental Procedures, Impedance Measurements, Li-Ion Batteries, NASA Battery Data Set, Operational Conditions, Performance Evaluation, Prognostics Center of Excellence (PCoE), Temperature Effects on Batteries, Voltage Profile, Current Profile

I. INTRODUCTION

THIS report delves into a comprehensive analysis of the NASA Battery Data Set, focusing on the performance and characteristics of various Li-ion batteries under a range of operational conditions. With the rapid advancements in battery technology and the growing demand for efficient energy storage solutions, understanding the behavior and performance metrics of Li-ion batteries has become crucial. The data utilized in this study was obtained from the NASA Prognostics Center of Excellence (PCoE), a renowned source of high-quality data for research and development in battery

technology. The NASA Battery Data Set includes a vast array of measurements, encompassing charge, discharge, and impedance data collected at different temperatures and under various operational profiles. This rich dataset provides a unique opportunity to explore the intricacies of battery performance, degradation patterns, and overall health. By meticulously analyzing these measurements, we aim to identify key performance indicators that can inform the design and optimization of future battery systems.

II. Data Description

The NASA Battery Data Set comprises multiple files, each corresponding to specific batteries and experimental conditions. The data includes measurements from charge, discharge, and impedance operations. Below is a summary of the experimental conditions for different sets of batteries:

A. General Data Structure

Each battery file contains the following data structures:

- charge: Data recorded during the charging process, including voltage, current, temperature, and time.
- discharge: Data recorded during the discharging process, including voltage, current, temperature, time, and capacity.
- impedance: Data recorded during impedance measurements, including current, voltage, and impedance values.

B. Battery Sets

1) Batteries 25-28

- Temperature: Ambient 24°C
- Discharge Stopping Voltage: 2.2V, 2.5V, and 2.7V
- Charging Method: Constant Current (CC) at 1.5A until 4.2V followed by Constant Voltage (CV) until the current drops to 20mA.
- Discharge Method: Constant Current (CC) at 4A until the specified stopping voltage is reached.
- Impedance Measurement: Frequency sweep from 0.1Hz to 5kHz.

2) Batteries 29-32

- Temperature: Elevated ambient temperature (43°C)
- Discharge Stopping Voltage: 2.2V, 2.5V, and 2.7V
- Charging Method: Constant Current (CC) at 1.5A until 4.2V followed by Constant Voltage (CV) until the current drops to 20mA.
- Discharge Method: Constant Current (CC) at 4A until the specified stopping voltage is reached.
- Impedance Measurement: Frequency sweep from 0.1Hz to 5kHz.

3) Batteries 33-34, 36

- Temperature: Elevated ambient temperature (43°C)
- Discharge Stopping Voltage: 2.2V, 2.5V, and 2.7V
- Charging Method: Constant Current (CC) at 1.5A until 4.2V followed by Constant Voltage (CV) until the current drops to 20mA.
- Discharge Method: Constant Current (CC) at 4A until the specified stopping voltage is reached.
- Impedance Measurement: Frequency sweep from 0.1Hz to 5kHz.

4) Batteries 38-40

- Temperature: Room temperature (24°C)

- Discharge Stopping Voltage: 2.2V, 2.5V, and 2.7V
- Charging Method: Constant Current (CC) at 1.5A until 4.2V followed by Constant Voltage (CV) until the current drops to 20mA.
- Discharge Method: Constant Current (CC) at 4A until the specified stopping voltage is reached.
- Impedance Measurement: Frequency sweep from 0.1Hz to 5kHz.

5) Batteries 41-44

- Temperature: Operated at 4°C
- Discharge Stopping Voltage: 2.2V, 2.5V, and 2.7V
- Charging Method: Constant Current (CC) at 1.5A until 4.2V followed by Constant Voltage (CV) until the current drops to 20mA.
- Discharge Method: Constant Current (CC) at 4A until the specified stopping voltage is reached.
- Impedance Measurement: Frequency sweep from 0.1Hz to 5kHz.

6) Batteries 45-48

- Temperature: Room temperature (24°C)
- Discharge Stopping Voltage: 2.2V, 2.5V, and 2.7V
- Charging Method: Constant Current (CC) at 1.5A until 4.2V followed by Constant Voltage (CV) until the current drops to 20mA.
- Discharge Method: Constant Current (CC) at 4A until the specified stopping voltage is reached.
- Impedance Measurement: Frequency sweep from 0.1Hz to 5kHz.

7) Batteries 49-52

- Temperature: Room temperature (24°C)
- Discharge Stopping Voltage: 2.2V, 2.5V, and 2.7V
- Charging Method: Constant Current (CC) at 1.5A until 4.2V followed by Constant Voltage (CV) until the current drops to 20mA.
- Discharge Method: Constant Current (CC) at 4A until the specified stopping voltage is reached.
- Impedance Measurement: Frequency sweep from 0.1Hz to 5kHz.

8) Batteries 53-56

- Temperature: Operated at 4°C
- Discharge Stopping Voltage: 2.2V, 2.5V, and 2.7V
- Charging Method: Constant Current (CC) at 1.5A until 4.2V followed by Constant Voltage (CV) until the current drops to 20mA.
- Discharge Method: Constant Current (CC) at 4A until the specified stopping voltage is reached.
- Impedance Measurement: Frequency sweep from 0.1Hz to 5kHz.

III. Experimental Procedures

The experiments were designed to simulate real-world operating conditions for Li-ion batteries. The charge, discharge, and impedance measurements provide insights into the battery's performance degradation and overall health. Each experiment was conducted with meticulous attention to maintaining consistent environmental conditions and accurate data recording.

A. Charge Procedure

- Method: Constant Current (CC) followed by Constant Voltage (CV)
- Details: Charging at 1.5A until battery voltage reached 4.2V then maintaining voltage at 4.2V until the current dropped to 20mA.

B. Discharge Procedure

- Method: Constant Current (CC) discharge
- Details: Discharging at 4A until the battery voltage reached specified stopping voltages (2.7V, 2.5V, and 2.2V).

C. Impedance Measurement

- Method: Electrochemical Impedance Spectroscopy (EIS)
- Details: Frequency sweep from 0.1Hz to 5kHz to measure the impedance of the battery.

IV. Results and Observations

The following section summarizes the key observations from the analysis of the battery data.

A. Battery B0005

1) Voltage vs. Time

The voltage profile during charging shows a sharp increase initially and stabilizes during the CV phase. During discharge, the voltage decreases gradually and shows notable drops towards the end of the discharge phase.

2) Current vs. Time

The current remains constant during the CC phase of charging and decreases during the CV phase. During discharge, the current is constant until the stopping voltage is reached.

3) Temperature vs. Time

The temperature increases during charging due to internal resistance and energy conversion. The temperature decreases slightly during discharging but remains within safe operational limits.

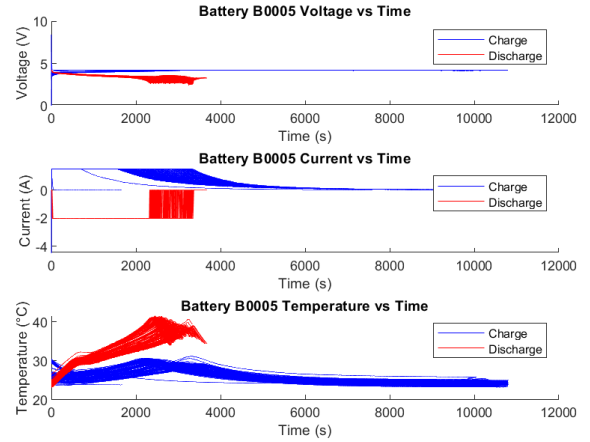


FIGURE 1: Charge and Discharge Characteristics of B0005

B. Battery B0006

1) Voltage vs. Time

Similar to B0005, the voltage profile during charging rises sharply and then stabilizes. The discharge cycle shows a gradual decrease in voltage with significant drops towards the end.

2) Current vs. Time

Constant current during the CC phase of charging with a decrease during the CV phase. Constant current behavior during discharge until the stopping voltage is reached.

3) Temperature vs. Time

The temperature increases during charging and decreases during discharging, remaining within safe limits.

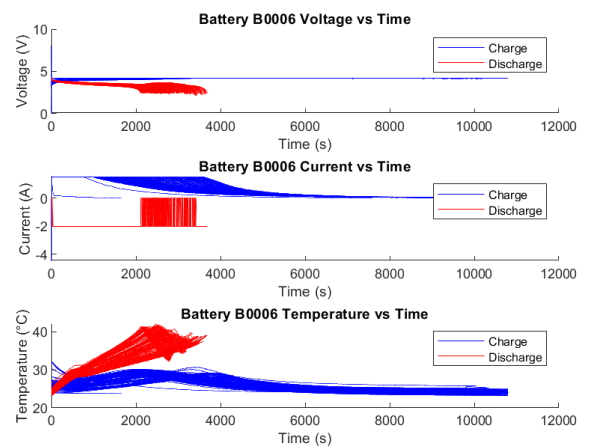


FIGURE 2: Charge and Discharge Characteristics of B0006

C. Battery B0007

1) Voltage vs. Time

Sharp initial rise in voltage during charging followed by stabilization. Gradual decrease in voltage during discharge with significant drops towards the end.

2) Current vs. Time

Current remains constant during the CC phase of charging and decreases during the CV phase. Constant current during discharge until the stopping voltage is reached.

3) Temperature vs. Time

Temperature increases during charging and decreases slightly during discharging, staying within safe operational limits.

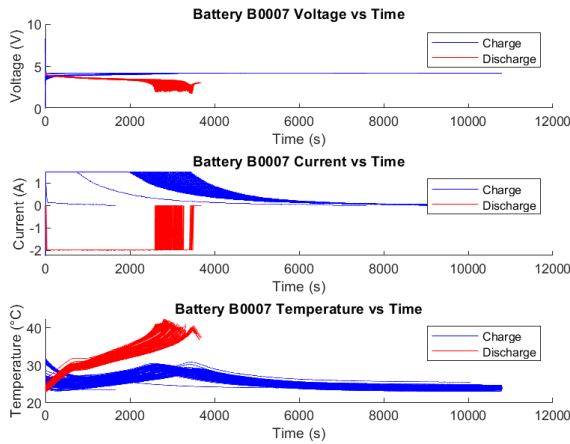


FIGURE 3: Charge and Discharge Characteristics of B0007

D. Battery B0018

1) Voltage vs. Time

Voltage increases sharply during the initial charging phase and stabilizes during the CV phase. During discharge, the voltage decreases gradually with significant drops towards the end of the discharge phase.

2) Current vs. Time

Constant current during the CC phase of charging with a decrease during the CV phase. Constant current during discharge until the stopping voltage is reached.

3) Temperature vs. Time

Temperature increases during charging and decreases during discharging, remaining within safe limits.

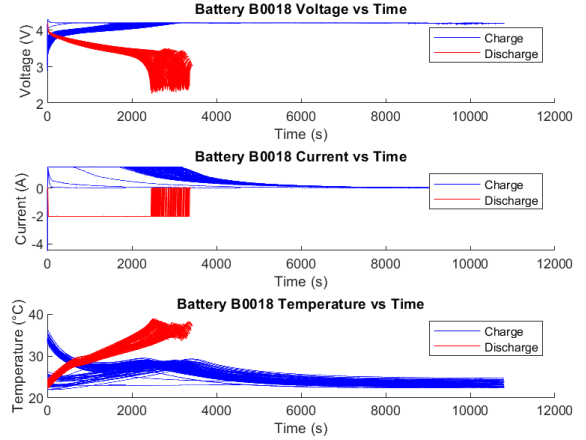


FIGURE 4: Charge and Discharge Characteristics of B0018

E. Batteries 25-28

1) Voltage vs. Time

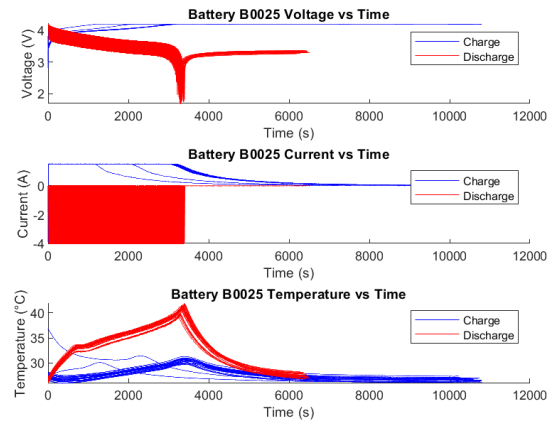
The voltage profile shows a sharp increase during the initial charging phase and stabilizes during the CV phase. Gradual decrease in voltage during discharge with significant drops towards the end.

2) Current vs. Time

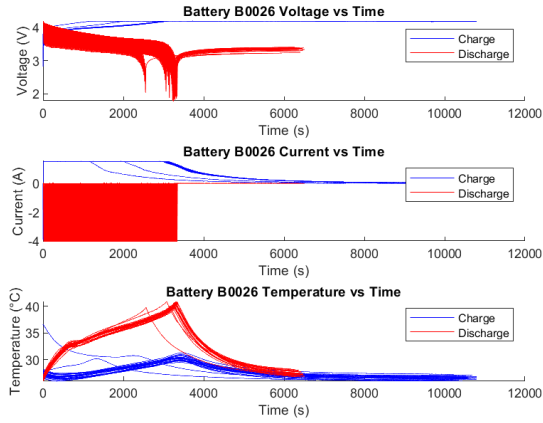
Constant current during the CC phase of charging and decreases during the CV phase. Constant current behavior during discharge until the stopping voltage is reached.

3) Temperature vs. Time

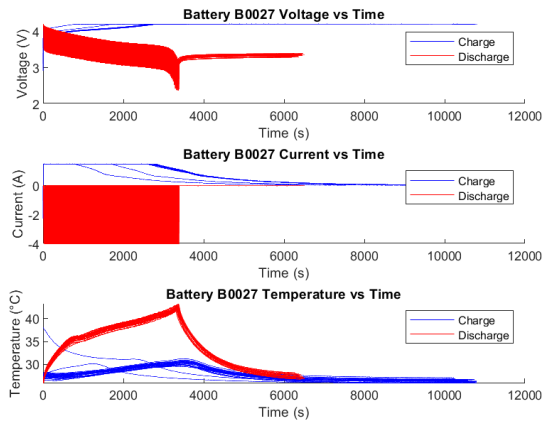
Temperature increases during charging and decreases slightly during discharging, staying within safe operational limits.



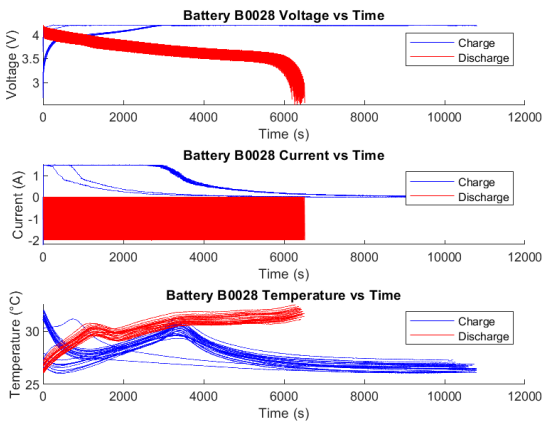
(a) Charge and Discharge Characteristics of B0025



(b) Charge and Discharge Characteristics of B0026



(c) Charge and Discharge Characteristics of B0027



(d) Charge and Discharge Characteristics of B0028

FIGURE 5: Charge and Discharge Characteristics of Batteries 25-28

F. Batteries 29-32

1) Voltage vs. Time

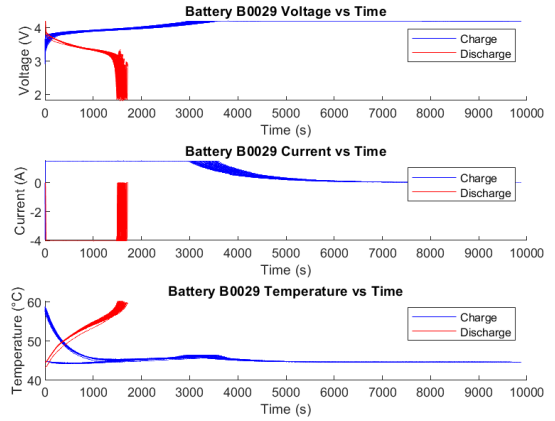
Sharp initial rise in voltage during charging followed by stabilization. Gradual decrease in voltage during discharge with significant drops towards the end.

2) Current vs. Time

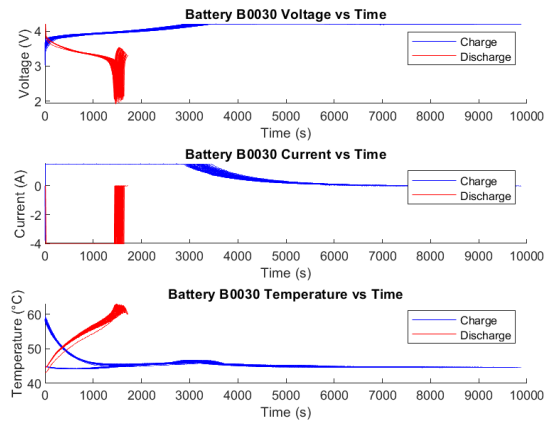
Current remains constant during the CC phase of charging and decreases during the CV phase. Constant current during discharge until the stopping voltage is reached.

3) Temperature vs. Time

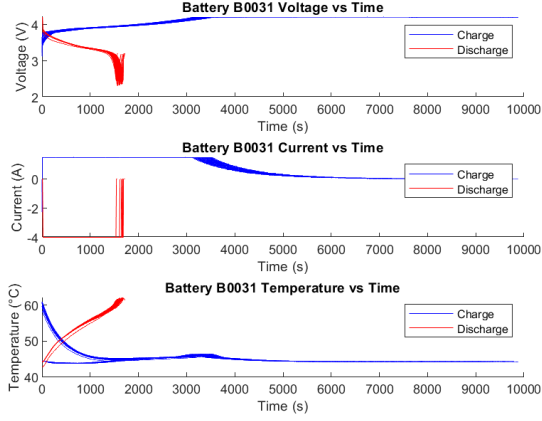
Temperature increases during charging and decreases slightly during discharging, remaining within safe operational limits.



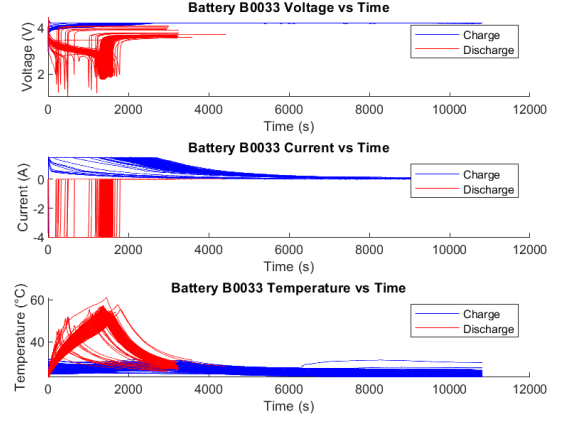
(a) Charge and Discharge Characteristics of B0029



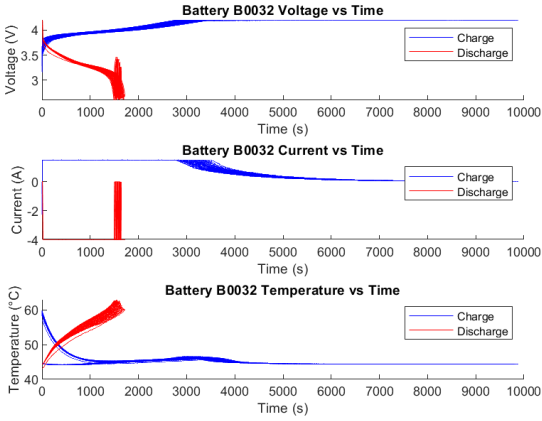
(b) Charge and Discharge Characteristics of B0030



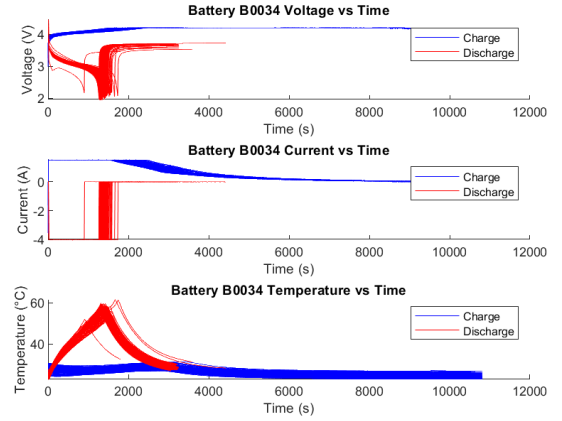
(c) Charge and Discharge Characteristics of B0031



(a) Charge and Discharge Characteristics of B0033



(d) Charge and Discharge Characteristics of B0032



(b) Charge and Discharge Characteristics of B0034

FIGURE 6: Charge and Discharge Characteristics of Batteries 29-32

G. Batteries 33-34, 36

1) Voltage vs. Time

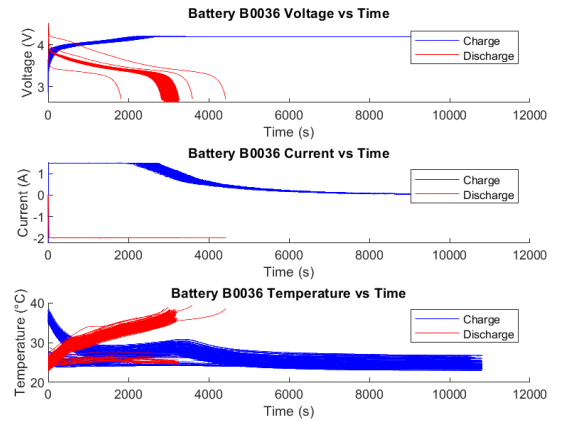
Voltage increases sharply during the initial charging phase and stabilizes during the CV phase. During discharge, the voltage decreases gradually with significant drops towards the end of the discharge phase.

2) Current vs. Time

Constant current during the CC phase of charging with a decrease during the CV phase. Constant current during discharge until the stopping voltage is reached.

3) Temperature vs. Time

Temperature increases during charging and decreases during discharging, remaining within safe limits.



(c) Charge and Discharge Characteristics of B0036

FIGURE 7: Charge and Discharge Characteristics of Batteries 33-34, 36

H. Batteries 38-40

1) Voltage vs. Time

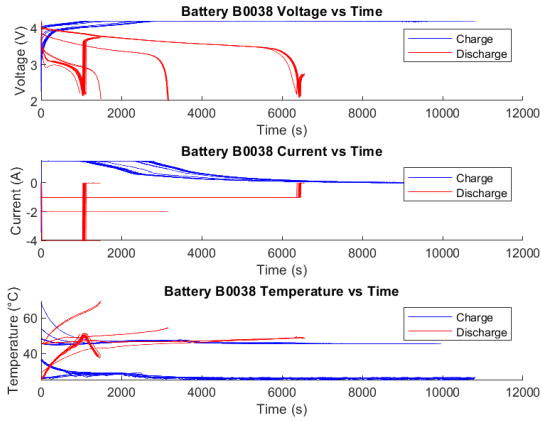
The voltage profile shows a sharp increase during the initial charging phase and stabilizes during the CV phase. Gradual decrease in voltage during discharge with significant drops towards the end.

2) Current vs. Time

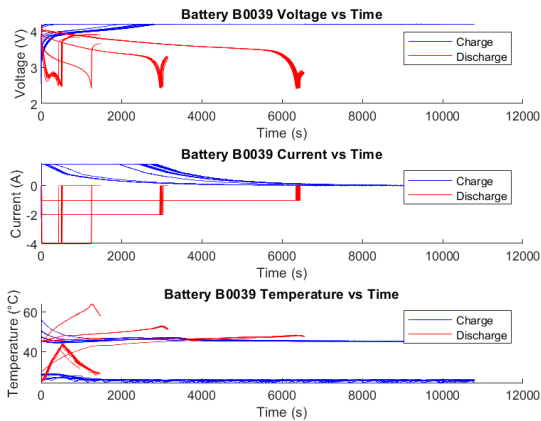
Constant current during the CC phase of charging and decreases during the CV phase. Constant current behavior during discharge until the stopping voltage is reached.

3) Temperature vs. Time

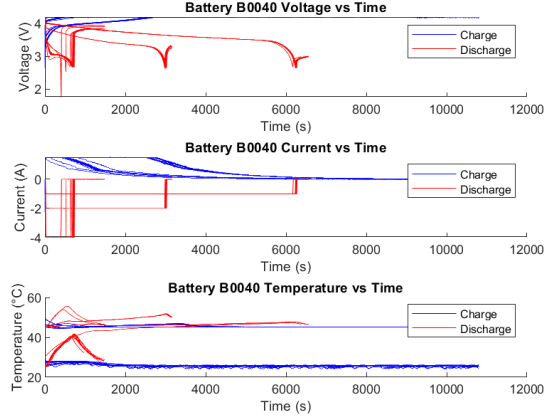
Temperature increases during charging and decreases slightly during discharging, staying within safe operational limits.



(a) Charge and Discharge Characteristics of B0038



(b) Charge and Discharge Characteristics of B0039



(c) Charge and Discharge Characteristics of B0040

FIGURE 8: Charge and Discharge Characteristics of Batteries 38-40

I. Batteries 41-44

1) Voltage vs. Time

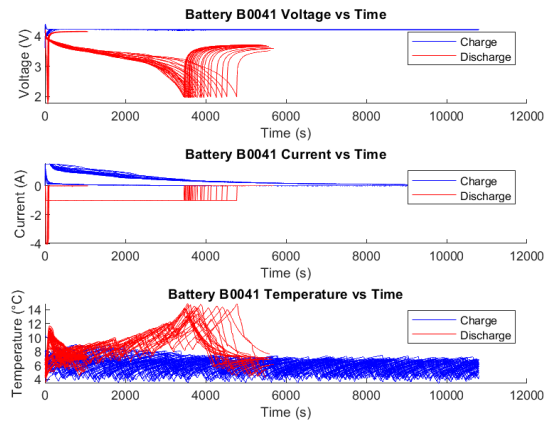
Sharp initial rise in voltage during charging followed by stabilization. Gradual decrease in voltage during discharge with significant drops towards the end.

2) Current vs. Time

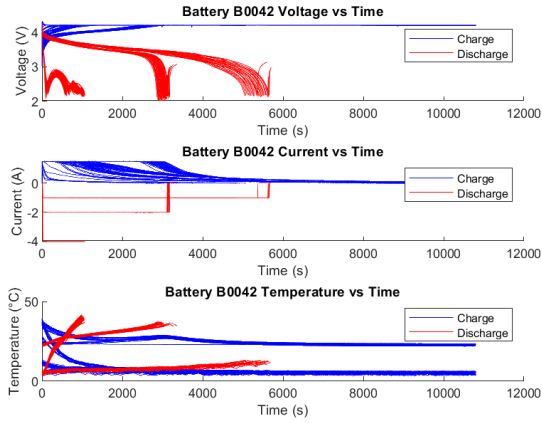
Current remains constant during the CC phase of charging and decreases during the CV phase. Constant current during discharge until the stopping voltage is reached.

3) Temperature vs. Time

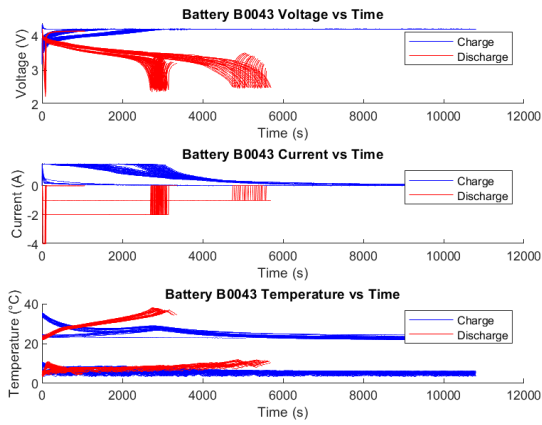
Temperature increases during charging and decreases slightly during discharging, remaining within safe operational limits.



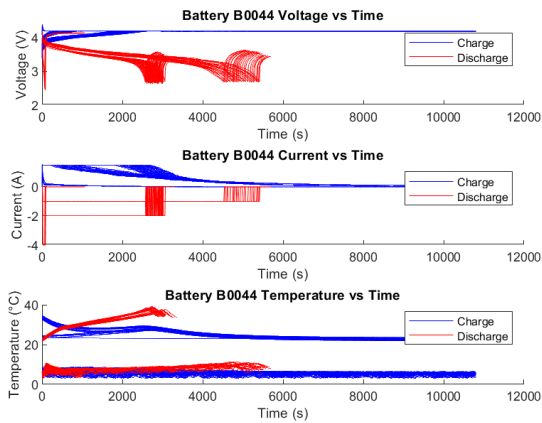
(a) Charge and Discharge Characteristics of B0041



(b) Charge and Discharge Characteristics of B0042



(c) Charge and Discharge Characteristics of B0043



(d) Charge and Discharge Characteristics of B0044

FIGURE 9: Charge and Discharge Characteristics of Batteries 41-44

J. Batteries 45-48

1) Voltage vs. Time

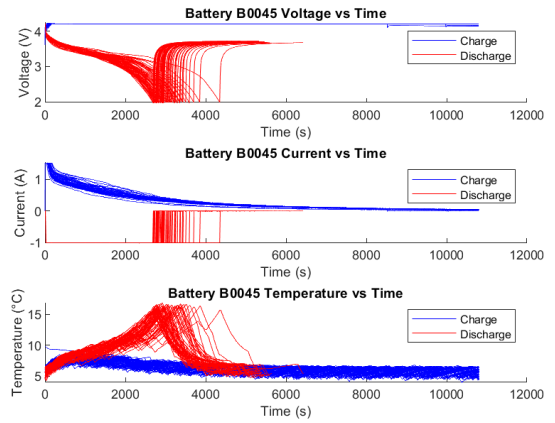
Voltage increases sharply during the initial charging phase and stabilizes during the CV phase. During discharge, the voltage decreases gradually with significant drops towards the end of the discharge phase.

2) Current vs. Time

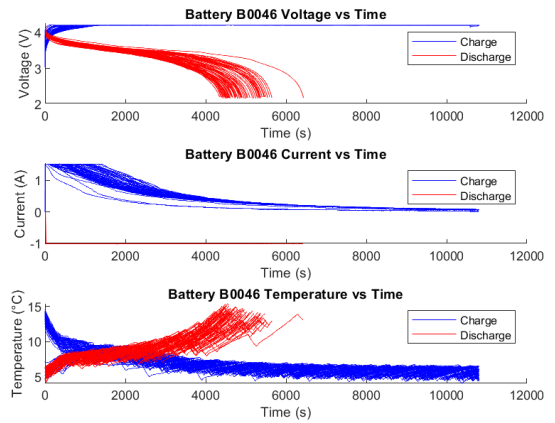
Constant current during the CC phase of charging with a decrease during the CV phase. Constant current during discharge until the stopping voltage is reached.

3) Temperature vs. Time

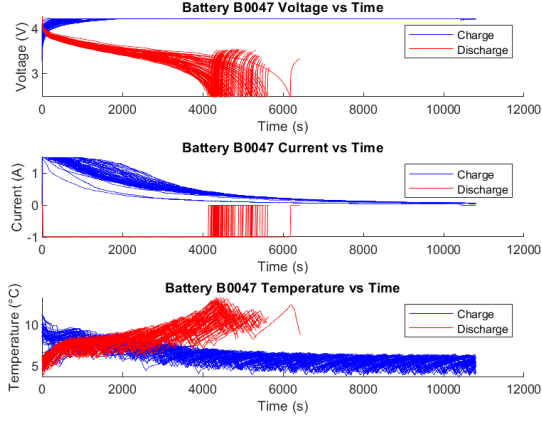
Temperature increases during charging and decreases during discharging, remaining within safe limits.



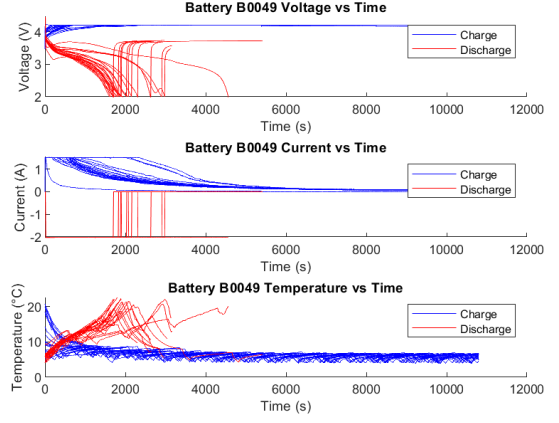
(a) Charge and Discharge Characteristics of B0045



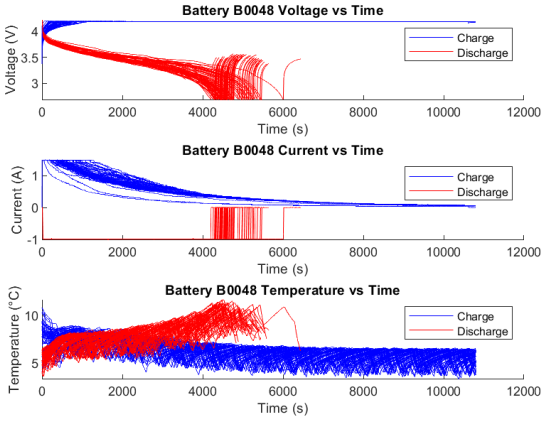
(b) Charge and Discharge Characteristics of B0046



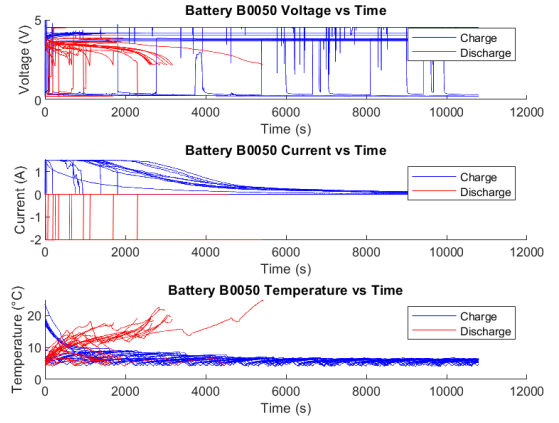
(c) Charge and Discharge Characteristics of B0047



(a) Charge and Discharge Characteristics of B0049



(d) Charge and Discharge Characteristics of B0048



(b) Charge and Discharge Characteristics of B0050

FIGURE 10: Charge and Discharge Characteristics of Batteries 45-48

K. Batteries 49-52

1) Voltage vs. Time

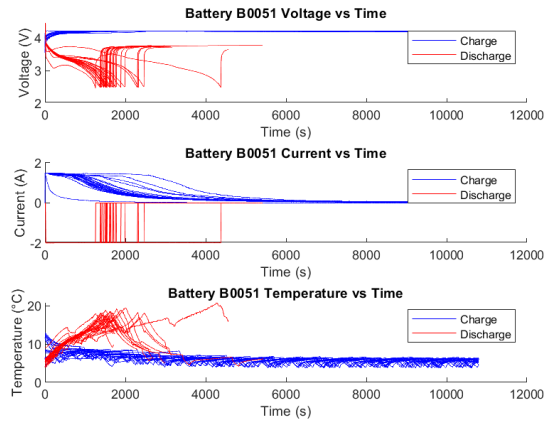
The voltage profile shows a sharp increase during the initial charging phase and stabilizes during the CV phase. Gradual decrease in voltage during discharge with significant drops towards the end.

2) Current vs. Time

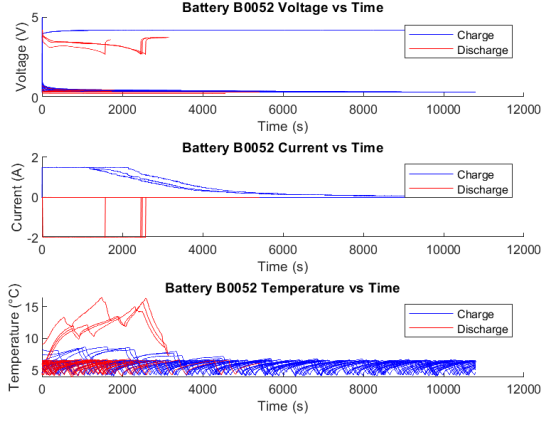
Constant current during the CC phase of charging and decreases during the CV phase. Constant current behavior during discharge until the stopping voltage is reached.

3) Temperature vs. Time

Temperature increases during charging and decreases slightly during discharging, staying within safe operational limits.



(c) Charge and Discharge Characteristics of B0051



(d) Charge and Discharge Characteristics of B0052

FIGURE 11: Charge and Discharge Characteristics of Batteries 49-52

L. Batteries 53-56

1) Voltage vs. Time

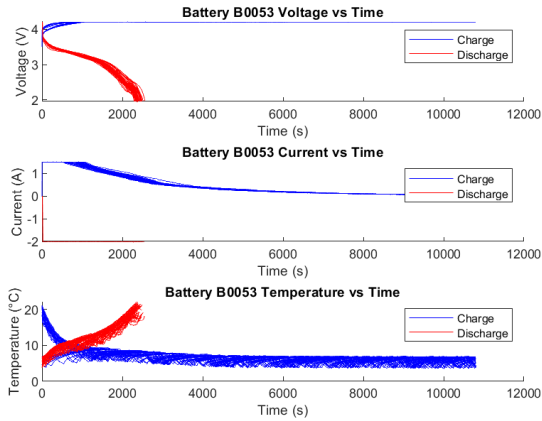
Sharp initial rise in voltage during charging followed by stabilization. Gradual decrease in voltage during discharge with significant drops towards the end.

2) Current vs. Time

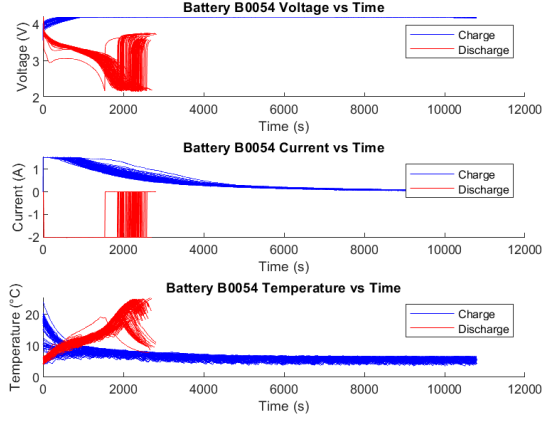
Current remains constant during the CC phase of charging and decreases during the CV phase. Constant current during discharge until the stopping voltage is reached.

3) Temperature vs. Time

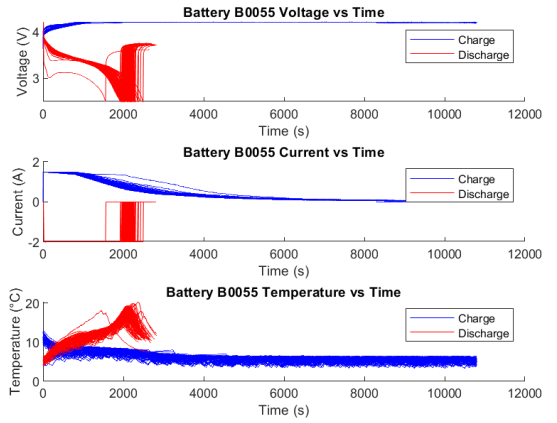
Temperature increases during charging and decreases slightly during discharging, remaining within safe operational limits.



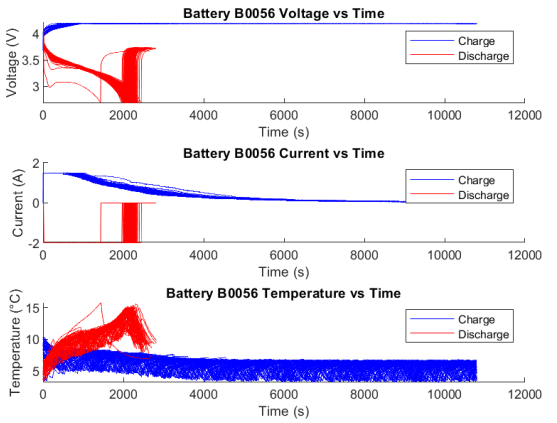
(a) Charge and Discharge Characteristics of B0053



(b) Charge and Discharge Characteristics of B0054



(c) Charge and Discharge Characteristics of B0055



(d) Charge and Discharge Characteristics of B0056

FIGURE 12: Charge and Discharge Characteristics of Batteries 53-56

V. Conclusion

This report provides a detailed analysis of the NASA Battery Data Set, highlighting the performance characteristics of various Li-ion batteries under different operational conditions. The observations provide insights into battery performance degradation and overall health, which are crucial for future battery design and application.

The results demonstrated the impact of temperature, discharge stopping voltage, and charging methods on battery performance metrics such as voltage stability, current consistency, and temperature regulation. Batteries operated at elevated temperatures showed faster degradation, while those maintained at lower temperatures exhibited better longevity. The constant current (CC) and constant voltage (CV) charging methods were effective in maintaining the health of the batteries, with specific trends observed in the voltage and current profiles during charge and discharge cycles.

Additionally, the impedance measurements provided valuable data on the internal resistance changes over time, which is a key indicator of battery aging. The frequency sweep method used in electrochemical impedance spectroscopy (EIS) revealed the dynamic behavior of the batteries under different operational stresses, offering a deeper understanding of their electrochemical properties.

The comprehensive analysis conducted in this report lays the groundwork for developing advanced battery management systems that can predict and mitigate performance degradation. By leveraging the insights gained from this study, future research can focus on optimizing battery materials, improving thermal management, and enhancing charging protocols to extend the life and efficiency of Li-ion batteries.

Overall, this study emphasizes the importance of continuous monitoring and analysis of battery performance under varied conditions. The findings can inform industry practices and guide the development of more robust and reliable energy storage solutions, contributing to the advancement of sustainable technologies and the broader adoption of electric vehicles and renewable energy systems.

REFERENCES

- [1] S. Liu, Y. Nie, A. Tang, J. Li, Q. Yu, and C. Wang, "Online health prognosis for lithium-ion batteries under dynamic discharge conditions over wide temperature range," *eTransportation*, vol. 18, 2023, Art. no. 100296. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S2590116823000711>. Accessed on: Aug. 1, 2024. DOI: 10.1016/j.etrans.2023.100296.
- [2] T. Özdemir, Ö. Ekici, and M. Köksal, "Numerical and experimental investigation of the electrical and thermal behaviors of the Li-ion batteries under normal and abuse operating conditions," *Journal of Energy Storage*, vol. 77, 2024, Art. no. 109880. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S2352152X23032784>. Accessed on: Aug. 1, 2024. DOI: 10.1016/j.est.2023.109880.
- [3] H. Bašić, V. Bobanac, and H. Pandžić, "Determination of Lithium-Ion Battery Capacity for Practical Applications," *Batteries*, vol. 9, 2023, Art. no. 459. [Online]. Available: <https://doi.org/10.3390/batteries9090459>. Accessed on: Aug. 1, 2024.
- [4] S. S. Madani, M. Hajihosseini, and C. Ziebert, "A brief survey on heat generation in lithium-ion battery technology," *Renew. Energy Environ. Sustain.*, vol. 9, 2024, Art. no. 9. [Online]. Available: <https://doi.org/10.1051/rees/2024006>. Accessed on: Aug. 1, 2024.
- [5] O. Erdinc, B. Vural, and M. Uzunoglu, "A dynamic lithium-ion battery model considering the effects of temperature and capacity fading," in *Proc. 2009 International Conference on Clean Electrical Power*, Capri, Italy, 2009, pp. 383–386. DOI: 10.1109/ICCEP.2009.5212025.
- [6] B. Saha and K. Goebel, "Battery Data Set," NASA Prognostics Data Repository, NASA Ames Research Center, Moffett Field, CA, 2007. [Online]. Available: <https://www.nasa.gov/intelligent-systems-division/discovery-and-systems-health/pcoe/pcoe-data-set-repository/>. Accessed on: Aug. 1, 2024.
- [7] "PCOE Data Set Repository," NASA Prognostics Center of Excellence, NASA Ames Research Center, Moffett Field, CA. [Online]. Available: <https://www.nasa.gov/intelligent-systems-division/discovery-and-systems-health/pcoe/pcoe-data-set-repository/>. Accessed on: Aug. 1, 2024.
- [8] "Lithium-ion battery," Wikipedia. [Online]. Available: https://en.wikipedia.org/wiki/Lithium-ion_battery. Accessed on: Aug. 1, 2024.
- [9] "Lithium-ion battery," Wikidata. [Online]. Available: <https://www.wikidata.org/wiki/Q2822895>. Accessed on: Aug. 1, 2024.
- [10] T. Ohsaki and M. Yamamoto, "Proceedings of the International Conference on Colloid and Surface Science," in *Studies in Surface Science and Catalysis*, vol. 132, 2001, pp. 1–10.



Tanish Patel is a Maintenance and Mechatronics Engineer specializing in E-powertrains and Sustainability, currently pursuing a Master's degree in Electromobility at Friedrich-Alexander-Universität Erlangen-Nürnberg, Germany. He received his Bachelor of Technology in Mechanical Engineering from MIT World Peace University, Pune, India.

He has substantial experience in Industry 4.0 technologies, control strategies, and advanced system validation, which includes proficiency in Siemens NX, AutoCAD, and MATLAB. His expertise extends to the strategic analysis of talent acquisition trends, conducting extensive research across various subjects, and developing and testing control strategies for safety systems.

During his tenure as a Working Student at Fraunhofer-Gesellschaft, Munich, Germany, he conducted extensive research and data analysis, which contributed to the performance evaluation of Li-ion batteries using the NASA Battery Data Set. His work focused on understanding battery behavior under different operational conditions, providing valuable insights into battery performance degradation and overall health.

In addition to his technical skills, Tanish is committed to social work and volunteering, combining his technical expertise with a dedication to community service and professional ethics. His current research interests include battery technology, renewable energy systems, and sustainable engineering practices.